

ARJUN PANKAJ JAISWAL

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EDUCATION

RCOEM, Nagpur – B.Tech in Information Technology | CGPA: 8.34 | Nagpur, India | 2023 – 2027

SKILLS

Languages: Java (primary), SQL, JavaScript

Frontend: HTML, CSS

Backend: Spring Boot, Spring Core, Spring Security, Spring Data JPA, Hibernate, JDBC, J2EE, REST APIs, JWT

Databases: MySQL, MongoDB

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), SOLID, Operating Systems, Database Management Systems (DBMS), Computer Networks, Multithreading & Concurrency

Design Patterns: DAO, MVC, Builder, Singleton, Adapter

Testing & Tools: JUnit 5, Git, GitHub, Maven, Gradle, Docker, Postman, JitPack

PROJECTS

SQL QueryBuilder | Java 17, JDBC, MySQL, Maven, JUnit 5, JitPack | Personal Project

GitHub: github.com/arjunpjaishwal/sql-querybuilder-lib

- Designed and built a Java library implementing the Builder Design Pattern to construct complex SQL queries through a fluent, chainable API, eliminating raw string concatenation and SQL injection risk.
- Engineered ACID-compliant transaction handling for multi-step operations (stock validation, order insertion, inventory update) with automatic rollback on failure.
- Built a complete JDBC execution layer (PreparedStatement, ResultSet, ResultSetMetaData) and a Singleton-based connection manager with double-checked locking for thread-safe initialization.
- Implemented a DAO layer abstracting all database access into simple entity-level methods (e.g., findByCity, findByAgeRange) used across User, Product, and Order entities.
- Published as an installable dependency via JitPack (sql-querybuilder-core:1.0.1) using a two-module Maven architecture: a core library and a separate runnable demo that consumes it exactly as an external project would.
- Applied all five SOLID principles with interface segregation between query (IQueryBuilder) and mutation (IMutationBuilder) builders; validated with 43 JUnit 5 unit tests requiring zero database dependency.

Mini Search Engine | Spring Boot, Java 17, MySQL, Jsoup, Groq API | Personal Project

GitHub: github.com/arjunpjaishwal/GSearch

- Built a multithreaded BFS web crawler using Jsoup and a 10-thread ExecutorService pool, indexing 500+ pages with thread-safe deduplication.
- Designed and implemented a custom inverted index with TF-IDF relevance ranking in MySQL, with no Elasticsearch/Lucene dependency.
- Built a paginated REST search API with multi-word query scoring and FetchType.LAZY optimization to minimize DB queries.
- Integrated Groq (Llama 3.1) for RAG-style AI summaries, grounded in retrieved search results to prevent hallucination.

Quora-Style Q&A Backend | Spring Boot, Spring Security, JWT, MySQL, Gradle, Postman | Personal Project

GitHub: github.com/arjunpjaishwal/quora-api

- Built a Quora-style REST API: users, follows, tagged questions, answers, nested comments and likes (Spring Boot, JPA, MySQL).
- Implemented stateless JWT authentication: BCrypt password hashing, a custom OncePerRequestFilter that validates signed tokens, and clean 401 responses for tampered, expired or malformed tokens.
- Enforced authorization: the user always comes from the token, never the request body; ownership checks (403) on edits, unique constraints against duplicate follows and likes, and DTOs that keep password hashes and other users' emails private.
- Built a paginated follow-based feed and text/tag search; verified every endpoint with a 62-request Postman collection.

EXPERIENCE

Training & Placement Coordinator – IT Dept | Training & Placement Cell, RCOEM | Aug 2024 – Present

- Bridged communication between students and CDPC for placements, training sessions, and recruitment drives.
- Guided students on recruitment preparation and campus placement processes; collected feedback on syllabus, aptitude prep, and learning preferences to improve training programs.
- Maintained student records and streamlined placement coordination workflows.

ACHIEVEMENTS

- Design Patent Granted – Smart Water Bottle focused on improved usability and smart functionality.
- Merit Award (Technology Startup) – RCOEM TBI & E-Cell Ideathon: recognized for a high-impact, scalable tech startup concept.
- 3rd Place – TechSprint Hackathon by Google Developer Group (GDG).
- LeetCode – 370+ problems solved in Data Structures, Algorithms, and Problem Solving.